Article published online: 2023-07-06

© 2023 IMIA and Georg Thieme Verlag KG

Health Information Exchange: Understanding the Policy Landscape and Future of Data Interoperability

A Jay Holmgren[1], Moritz Esdar[2], Jens Hüsers[2], João Coutinho-Almeida[3]
[1] University of California, San Francisco, USA
[2] University of Applied Sciences Osnabrueck, Germany
[3] Porto University, Portugal

Summary

Objectives: To review recent literature on health information exchange (HIE), focusing on the policy approach of five case study nations: the United States of America, the United Kingdom, Germany, Israel, and Portugal, as well as synthesize lessons learned across countries and provide recommendations for future research.

Methods: A narrative review of each nation's HIE policy frameworks, current state, and future HIE strategy.

Results: Key themes that emerged include the importance of both central decision-making as well as local innovation, the multiple and complex challenges of broad HIE adoption, and the varying role of HIE across different national health system structures.

Conclusion: HIE is an increasingly important capability and policy priority as electronic health record (EHR) adoption becomes more common and care delivery is increasingly digitized. While all five case study nations have adopted some level of HIE, there are significant differences across their level of data sharing infrastructure and maturity, and each nation took a different policy approach. While identifying generalizable strategies across disparate international systems is challenging, there are several common themes across successful HIE policy frameworks, such as the importance of central government prioritization of data sharing. Finally, we make several recommendations for future research to expand the breadth and depth of the literature on HIE and guide future decision-making by policymakers and practitioners.

Keywords

Health Information Exchange, interoperability, data sharing, health information technology, electronic health records

Yearb Med Inform 2023:184-94
http://dx.doi.org/10.1055/s-0043-1768719

1 Introduction

As health care has modernized in the 21st century, moving away from paper-based record keeping to digital workflows following the broad adoption of electronic health records (EHRs), health information exchange (HIE) and data interoperability – the process of electronically sharing data across unaffiliated organizations including care delivery providers, payers, public health agencies, and more - have become increasingly crucial components of a modern health system [1, 2]. Not only is widespread HIE critical to providing clinicians with a full picture of patient health status at the point of care to ensure quality and safety, robust interoperability could also slow medical cost growth through a reduction in duplicative utilization [3], and may reduce administrative burden on patients by letting their data follow them seamlessly across the continuum of care [4]. Further, the COVID-19 pandemic has illustrated the importance of readily-available patient and population-level health data aggregated across care delivery organizations – one of several critical HIE use cases [5]. Building a robust interoperable health system is therefore an important policy goal, and is essential to realizing the value of the significant investment made in digitizing health care delivery over the past several decades.

HIE appears simple in concept – the idea that all health data should be readily available to the patient and clinicians at the point of care, regardless of where that data was generated. However, the implementation of large-scale, robust HIE infrastructure and capabilities has proven difficult in health systems across the world [6, 7]. Building connectivity between unaffiliated health care organizations presents a wide array of challenges, ranging from technical issues around data standards, governance problems with what data is shared and who owns it, privacy and security concerns for both patients and organizations, costs of technology and implementation, competitive disincentives to allow data to flow easily to other organizations, and workflow questions on how to best present busy clinicians with actionable knowledge without burdening them with irrelevant information. These challenges are situated within the broader policy framework of each nation, both their specific approach to HIE as well as how they organize care delivery in general. For example, in some nations HIE faces financial disincentives as health care organizations are reluctant to share patient data that they see as a key strategic asset in a competitive market [8, 9]. Additionally, while the theoretical benefits of HIE are obvious, demonstrating value to patients, clinicians, and health systems at scale has been more difficult. There remain critical unanswered questions around HIE and interoperable data exchange, including how different public policy approaches have resulted in varying levels of HIE adoption and use, best practices for integrating HIE into clinical, reporting and administrative workflows, and identifying where HIE can generate the most value across a range of clinical, financial, and administrative scenarios.

In this survey paper, we discuss the concept and types of HIE, the benefits and challenges of implementing HIE systems, and use national case studies to highlight the diverse set of approaches to HIE from five different countries. We focus our review specifically on examining the policy frameworks employed by our case study nations around data sharing between acute care delivery organizations, and how they have influenced the development of HIE in each country. We then synthesize the current literature on HIE and provide recommendations for future work. Focusing on recent literature, we provide readers with a look at the current state of global research relating to HIE and identify opportunities for the informatics community to advance our knowledge and understanding of HIE across a range of domains.

2 Methods

We conducted a narrative review of the HIE and interoperability literature in both the peer reviewed and grey literature, focusing on highlighting the history, current state, and future strategy of five case study nations: the United States of America, the United Kingdom, Germany, Israel, and Portugal. We specifically focused on identifying literature from the last three years but did not exclude older or non-peer reviewed articles which provided important context or the most up-to-date information. The search strategy was broken down into three steps. First, we combined relevant search terms for identifying current research articles in the respective countries in MEDLINE via PubMed and Google Scholar:

> *(health OR healthcare OR clinical OR medical) AND (information exchange OR data sharing OR data exchange OR information sharing OR interoperability) AND (United States OR Germany OR United Kingdom OR England OR NHS OR Israel OR Portugal)*

Second, we researched the official websites of the respective health authorities for additional resources (e.g., strategy papers, government reports, policy documents or other information on current legislation). Third, we complemented the search with additional grey literature based on leads from the sources found in the first two steps. Furthermore, to contrast the state of HIE between the selected countries along comparable dimensions, we used the identified literature alongside complementary sources to classify: 1) the level of EHR adoption in acute care organizations (as an essential prerequisite for HIE); 2) the overall HIE maturity; 3) the level of HIE centralization; and 4) level of incentivization for HIE as either "low", "moderate", or "high" for each country (Table 1). We then summarized our findings across these international contexts to create a synthesized set of lessons learned, as well as areas for future study and research.

3 Results

3.1 Health Information Exchange – The Concept, The Verb, the Noun

As discussed in the introduction, the goal of HIE sounds simple – all health information should be available to the appropriate users (patients, clinicians, public health agencies, etc.) whenever necessary, regardless of where that data was generated. However, operationalizing this idea requires consideration of many socio-technical issues, e.g., What data should be shared with whom? What specific data elements? Who should access the data? How should data be accessed? Should the patients opt-in to data sharing, or should they be enrolled by default with the ability to opt-out? What standards and processes should be used to capture and transfer data and to ensure data privacy and security? These difficult questions, and many more, illustrate the complexity in turning HIE from an idea into a reality. HIE can therefore represent a wide array of electronic data sharing – ranging from sending flat PDF files via secure email to semantic interoperability, where standardized, structured, machine-readable data elements are transferred and integrated directly into the receiving organization's EHR without manual intervention [10, 11]. HIE can be a "push", where a health system or provider sends patient data to another provider during a transition of care, or a "pull", where the receiving system queries for any patient data available from other providers.

The term "HIE" is frequently used as both a verb and a noun [12, 13]. HIE the *verb* refers to the act of data transfer, sharing data between two health care organizations, of which there are many possible technical approaches and mechanisms of data exchange. HIE the *noun*, however, most frequently refers to an organization that is facilitating the data transfer, sometimes known as health information exchange organizations (HIOs). They are most often vendor-neutral, compared to the vendor-mediated tools that connect organizations using the same EHR vendor [14], and HIOs provide technical capability and governance frameworks for data exchange [15, 16]. In the United States

Table 1 Dimensions of Health Information Exchange Across Five Nations.

	Electronic Health Record Adoption in Acute Care Organizations	Overall Health Information Exchange Maturity	Level of Health Information Exchange Centralization	Incentives for Health Information Exchange
United States	High [21]	Moderate [27]	Low [14]	Moderate [27]
Germany	Moderate [52,104]	Low [49,52]	High [17]	Moderate [105]
The United Kingdom	High [58,106]	Moderate to High [7,61,107]	High [7,55]	Moderate to High [7,108]
Israel	High [72,109,110]	High [6,71,111]	Moderate [54,71,112]	High [71,112,113]
Portugal	High [87,114]	Moderate to High [79,80,82]	Moderate to High [81,82]	Moderate to High [83,84,93]

(US), for example, these organizations can be non-profit or for-profit, based in a specific geography (e.g., state or local) or national, and may be based on a specific framework designed to streamline data exchange across different platforms and technology vendors such as Carequality, and some HIEs are a "network-of-networks" that link several other HIEs together, while in Germany the gematik, a national eHealth organization, provides similar services but with a much broader mandate to set policy, standards, and governance relative to regional US HIOs [17, 18].

Finally, the terms HIE and interoperability are often used interchangeably. However, they are distinct concepts – HIE involves any health data transfer, in any format, whereas interoperability refers specifically to the exchange of structured data elements. In this way, sending scanned paper files in PDF form electronically would represent HIE, but not interoperability, whereas data that was sent in a machine readable format and integrated into the receiving provider's EHR would be described as interoperability [19].

3.2 The State of International Health Information Exchange: a study of five case nations

3.2.1 The United States of America

Following the passage of the Health Information Technology for Economic and Clinical Health (HITECH) Act in 2009, acute care hospitals and primary care physician offices in the U.S. rapidly adopted EHRs in response to federal incentives included in the Meaningful Use (MU) program [1, 20, 21]. While not as directly incentivized with payments to the same level as EHR adoption, a variety of public policies have focused on encouraging interoperability and HIE within US health care delivery organizations. Most directly, HITECH included several subsidies and grant programs to fund the creation, expansion, and operation of HIOs, often operating at the state or local level, to build HIE infrastructure. Several MU requirements in the later stages of the program directly incentivized at least some level of data sharing, such as requiring a certain proportion of patient care transitions to outside organizations have summary of care records sent electronically, and the EHR certification program to attest to MU ensured that EHR products were able to send and receive data electronically. These requirements have been expanded by the Centers for Medicare and Medicaid Services (CMS) Promoting Interoperability program [22]. Several policy initiatives have attempted to promote HIE adoption and use through aligning the financial incentives of care delivery organizations with data sharing, such as the proliferation of value-based payment models including bundled payments or Accountable Care Organizations (ACOs) that reward providers for reducing population-level utilization and spending [23]. These types of value-based payment reforms were intended to overcome the misaligned financial incentives in the fee-for-service payment model for care delivery in the US, and which provides no financial incentive to use HIE to avoid redundant or unnecessary services [19, 24].

Despite the wide range of policy incentives targeted at encouraging HIE, the state of interoperability (within and across the health care delivery eco-system) in the US varies. The US Office of the National Coordinator for Health Information Technology (ONC) tracks hospital interoperability using four domains of: 1) finding/querying for data; 2) sending data electronically; 3) receiving data electronically; and 4) integrating outside data into the EHR without manual intervention [25, 26]. Reflecting the structure of the MU certification requirements, as of 2018, the ability to send data electronically is nearly ubiquitous, with nearly 90% of hospitals reporting they often or routinely send patient summary of care records electronically. In contrast, fewer than two-thirds of US hospitals report being able to integrate outside data into their EHR, and less than half (45.4%) reported engaging in all four ONC's interoperability domains [27]. Research has also found that hospitals and public health agencies lack bi-directional interoperability, which caused significant issues aggregating population-level data in the early days of the COVID-19 pandemic [5, 28, 29]. While interoperability has made progress over the past several years [30, 31], it has been significantly slower than EHR adoption [1, 21]. Ambulatory physician offices report even lower levels of interoperability or HIE engagement, as they often lack the organizational resources such as a full-time information technology staff to build and maintain HIE connections [32], though national-level data for these organizations is less detailed [33].

The current state of HIE and interoperability in the US is a patchwork of connectivity without a centralized national HIE approach, and whether a patient's data follows them as they transition across care providers is determined more by whether their organizations are on the same EHR vendor or participate in the same method of HIE. Significant heterogeneity in state policy for HIE, including different approaches to governance, financial support, data privacy and patient consent laws (opt-in vs opt-out), also contributes to variation in interoperable data exchange in the US [34, 35]. Many organizations report participating in multiple different modalities of sharing patient data, such as vendor (EHR)-based systems [36] as well as regional HIEs and national HIE networks [37]. Interoperability between acute care providers and other health care organizations, including long-term care facilities or public health agencies, is also limited and uneven across the country [5, 38]. These results are the reflection of the somewhat scattered policy environment, with few prescriptive requirements and many diffuse and weak incentives to share data without a comprehensive path towards national-level interoperability. The value-based payment models to align financial incentives for data sharing were only modestly successful, with ACO hospitals sharing data with more types of partners, but with a lower overall volume of records exchanged relative to non-ACO participants [39, 40], and the lack of HIE infrastructure has presented a significant barrier for ambulatory practices participating in population health management programs [41].

Future Considerations & Strategy: In 2016, Congress passed the 21st Century Cures Act (Cures), which included several provisions to improve patient access to data as well as interoperability between provider organizations. Cures provides a framework for HIE in the US, including provisions

such as outlawing "information blocking", the practice of intentionally and knowingly blocking patient data access by health information technology developers or health care provider organizations [8], as well as updates to the ONC EHR certification criteria to mandate the use of standardized application programming interfaces (APIs) that facilitate patient access to their own health data and HIE using a common set of standards known as Fast Healthcare Interoperability Resources (FHIR) [42]. Further, the act outlined the Trusted Exchange Framework and Common Agreement (TEFCA), a voluntary technology and governance model that seeks to streamline the exchange of patient data across the multiple fragmented HIO networks that exist in the US, reducing the need for provider organizations to participate in multiple HIE networks [43]. Rulemaking for the Cures Act was finalized in 2021, with information blocking and FHIR API requirements now in place, and the first several organizations designated as Qualified Health Information Networks (QHINs) in the TEFCA framework were named in early 2023 with plans to go live within the next year [44,45]. This more prescriptive framework represents a departure from previous efforts focused on aligning financial incentives, and will hopefully deliver significant improvements to interoperability in the US.

3.2.2 Germany

Germany initiated the development of a dedicated infrastructure for HIE, called the telematics infrastructure (TI), for the statutory health insurance (SHI) system in 2004. The TI uses an opt-in patient consent model that also requires the patients provide their clinician with a physical card that authorizes access and changes to the data stored in the TI. This was followed, however, by a prolonged timeframe of disagreements and mutual obstruction among the main shareholders of the national organization that was made responsible for the development of the TI (the so-called gematik) [46]. The ensuing stalemate was said to be driven by a mix of poor federal governance, lack of a strategy, lack of consensus-building, excessive data protection concerns on the part of the physician associations, and miss-aligned financial incentives which severely hampered progress towards widespread HIE practices [46–50]. From 2016 on, renewed legislative efforts were made to implement the TI and to subsequently specify how data ought to be exchanged through a central EHR system that utilizes the TI. Also, the federal ministry of health intervened in 2019 and took over the majority shares of the gematik to be able to undercut further deadlocks between the main parties (particularly the National Association of SHI Funds, the National Association of SHI Physicians, and the German Hospital Federation).

As part of the latest legislative initiatives, the National Association of SHI Physicians was made exclusively responsible for specifying syntactic and semantic standards for exchanging health information via the central EHR system. Although this recently resulted in an increased focus on international standards (e.g., FHIR, SNOMED CT, and LOINC), Germany's HIE capabilities remain at a low level. While all German EHRs are now required to be interoperable with the TI infrastructure [51], a survey from 2017 showed that only a fraction of German hospitals are able to regularly exchange data with other providers or the patients electronically [49]. More recent data corroborate these findings [52] – and note poor HIE between physicians in ambulatory care settings and low adoption rates of the central EHR by patients [17, 53]. However, the central EHR system and related components are still being developed and the current government has committed to switching from an opt-in to an opt-out principle for all members of the SHI, thus hoping to significantly increase HIE engagement. Still, Germany is generally considered to perform worse in HIE as compared to the US, England, Portugal and Israel [6, 54].

Future Considerations & Strategy: Germany's previous "hands-off" approach to facilitating HIE can hardly be classified as strategic. And with few exceptions, neither the health insurance companies nor the various providers or the state governments developed HIE systems themselves from the 'bottom-up'. To cope with the resulting inadequacies, the federal ministry of health started intervening through numerous legislative initiatives, especially after a change in leadership in 2018. These were primarily concerned with further developing the TI as a single, centralized HIE system and the central EHR system as well as a comprehensive financial incentive program for hospitals that, among others, aims at the widespread adoption of patient portals to exchange data with the patients and other providers. From an organizational viewpoint, clearer lines of responsibilities were instituted within the self-governing bodies (e.g., the physician's association was made solely responsible for defining standards to be used in the central EHR and the health insurance companies for providing the EHR to their beneficiaries). Additionally, a decree was issued in 2021 to create a governance structure for ensuring universal interoperability with the so-called Interop Council at the center and binding directives for providers and vendors to adhere to certain standards. However, some of these initiatives were criticized as being reactive rather than proactive, partly disconnected from one another and at risk of creating parallel HIE structures. Hence, calls for a coherent strategy followed, which the current federal ministry has taken up and is expected to announce in 2023.

3.2.3 The United Kingdom

The English National Health Service (NHS) has had a long and checkered history of building its HIE capabilities [55]. Capitalizing on its centralized make-up and control, the NHS has focused on defining and propagating technical and data standards as early as 1992. Significant investments were made early on which led to the incremental establishment of a connectivity framework, comprising the Health and Social Care Network, the "NHS Spine", as well as various other central and local services for enabling HIE [7, 55]. Current rules for data sharing in the United Kingdom (UK) use an implied consent (or explicit 'opt-out') model for direct patient care purposes [56]. The NHS's strategic approach has often been characterized as largely coming from the 'top-down' [55, 57]. Particularly the much-discussed National Programme for Information Technology (NPfIT) from 2003 exhibited high degrees of central control. It

attracted a lot of criticism and is said to have failed partly because of this high degree of centrality [57]. Still, the NPfIT contributed fundamentally to the connectivity framework in its current form. The NHS subsequently moved on to allow for more local variation in their following strategies, thus shifting towards what has been characterized as a 'middle-out' approach [55].

The NHS's early commitment to HIE, together with a convoluted interplay of policy and technology changes over time has resulted in a rather complex ecosystem of patient health records [58]. Perhaps most importantly, Summary Care Records (SCRs) containing basic patient information such as medications and allergies are available and used by almost all general practitioners via the NHS Spine digitally and this functionality is required for EHRs [59], and more comprehensive shared records are increasingly being adopted by local NHS organizations and systems. The increase in shared records partly results from more recent initiatives – particularly the Information Governance Framework for Integrated Health [60]. However, despite the NHS's relatively advanced stage of HIE adoption [7], data sharing and interoperability challenges remain – particularly with regard to the EHR vendor specificity of HIE systems in place as well as information exchange with and between hospitals [58, 61]. While a recent study by Watkins *et al.* shows that the use and user satisfaction with an EHR vendor-specific solution is quite high, another study by Warren *et al.* points to substantial misalignments and information exchange barriers that remain between hospital trusts that use different HIE solutions [58, 62].

Future Considerations & Strategy: The NHS continues to actively adjust and adapt its HIE strategy from a rather central angle while explicitly leaving room for variations by local NHS authorities and systems. Most recently, this finds expression in its 2022 "data saves lives" strategy and in the substantial reforms of its governance structure [63]. These activities might best be summarized as the ambition to consolidate, advance, and capitalize on its HIE capabilities. As a principal challenge in delivering universal interoperability still lies in the fragmentation of responsibilities for the HIE infrastructure [61], the NHS plans to consolidate the previously separated public bodies NHS Digital, NHSX and NHS Improvement under one roof in 2023 - that of NHS England to form a centralized, single HIE system. This consolidation is also reflected in the data strategy's premise itself, which views HIE not as an isolated matter, but rather as an integral part of a wider and more far-reaching data strategy. It thereby, among other things, aims at giving patients greater control over their data and creating secure data environments for research. From a technical viewpoint, the strategy commits to enhancing the usage of international standards (particularly in the form of UK-wide HL7 FHIR Profiles), the usage of cloud services, and services to centrally find and retrieve records from the various systems. In terms of advancements, it aims to extend the types of data exchanged such as health information from wearables and, crucially, social care data. Lastly, the strategy also sketches out further steps to better harness health data for research, promote the application of AI technologies in reference to the Topol review [64], and steps to continue to work towards establishing a learning health system – thus capitalizing on its current and anticipated HIE capabilities [65], These activities reflect an integrated, holistic policy approach and a structured governance framework to integrate care pathways across siloed responsibilities to encourage health information technology-based innovation [66].

3.2.4 Israel

HIE structures and practices in Israel have been closely tied to the structure of the country's healthcare system [67]. Within their universal national health insurance system, each Israeli citizen is a member of one of the four not-for-profit Health Maintenance Organizations through which they receive both insurance and care. While the Ministry of Health (MoH) is responsible for overarching policy, regulation, planning, budgeting, etc., the HMOs have been relatively independent in the way they design their healthcare services. The competition among them as well as their integrated accountability for the patients across various care settings is said to have facilitated innovation-seeking strategies from the bottom-up, including the adoption of HIE solutions within the HMOs [54, 67]. The largest of the four HMOs, 'Clalit' (which covers roughly 52% of the population and exhibits the highest degree of vertical integration from primary to tertiary care services), was an early adopter of an HIE system called OFEK – a distributed system that enabled the retrieval of basic patient information from any kind of local EHR system through a specific interface, with patients automatically enrolled with an opt-out option [68, 69]. Although the system was found to create local benefits in terms of reduced test ordering and nearly all Israeli EHRs were interoperable within HMOs [70], information exchange outside the HMOs remained insufficient and patient access to their medical data is still rather limited for many citizens [6]. In light of these limitations, the MoH started taking on a more active role to promote widespread HIE. This was first expressed by adopting OFEK as a single centralized HIE system to be used on a national level across all service providers by 2014 [70, 71]. Subsequently, the MoH also published a digital health strategy in 2018 in which HIE was recognized as a foundational strategic element [71, 72]. Correspondingly, OFEK was further developed and eventually upgraded to a new version with more comprehensive and advanced functionalities, called EITAN. The implementation of EITAN was supported by a financial incentive program for the adopting organizations and designed as an opt-out system to ensure widespread use [71, 73].

Most Israeli healthcare organizations have now adopted EITAN. Similar to its predecessor OFEK, EITAN uses a decentralized architecture: clinical data are stored in local databases within the provider organizations based on a nationally consented clinical dataset. These databases operate using a proprietary and detailed mapping of clinical data elements into a semantic interoperability standard, and receive their data from the organization's EHR and are connected through a national HIE network which has its hub at the MoH [74]. Thus, patient data can be shared and accessed by authorized personnel in all provider organizations. However, due to the usage of varying HIE

solutions in conjunction with different local EHR systems over time, challenges remain concerning the consistent use of standards and terminologies [71, 75]. Furthermore, Itzhaki *et al.* recently found that many ward nurses, contact nurses (who are responsible for care coordination across providers), and patients are still unaware of EITAN which impedes its effective use [76]. Additionally, there is a nascent but growing community emphasizing the use of HL7 FHIR standards in Israel, including the MoH, which launched the first FHIR-based projects in 2021 [75]. But despite these drawbacks, Israel's HIE capabilities are viewed as rather advanced as compared to many other developed countries [54, 72].

Future Considerations & Strategy: The MoH continues to actively support the adoption and further development of its nationwide HIE solution EITAN based on a public-private partnership [72, 77]. Thus, local-level HIE innovations within the HMOs have largely given way to more centralized strategic planning and execution by the MoH to promote coherence across health system silos. For example, this is currently reflected in efforts to improve the interoperability and accessibility of clinical information for advanced data analytic services by pursuing the widespread implementation of HL7 FHIR [75]. Despite this tendency for more centralized planning in many regards, the MoH still aims to facilitate bottom-up innovation based on their past successes by, for example, providing funding for digital health services such as third-party clinical decision support tools that utilize data from the HIE network [72, 78]. Going forward, it is also expected to aim for enhanced patient access to their medical data through mobile, personal health record solutions [71] – thereby making further progress in realizing the paradigm shift towards more personalized medicine.

3.2.5 Portugal

Portugal's national HIE strategy has gone through several transitions since its foundation in the 1990s. If in the late 90s and early 2000s, the Portugal eHealth ecosystem was recognized as being the most advanced for its time, i.e. being one of the first countries to implement nationwide unique health identification, it has been lacking the same propelling drive since the early 2000s. The prevalent idea until the early 2000s was monolithic solutions that, in theory, would cover most of the institutions' Health Informatics necessities. However, the 2010s brought a different perspective to the Portuguese landscape.

This happened for mainly two reasons, SONHO as a mandatory software and LIGHt (Local Interoperability Gateway for Healthcare). SONHO is a public admission-discharge-transfer (ADT) health information system (HIS) and in 2013 all public hospitals were forced by law to migrate their current solutions to it [79]. This software is the base for registering administrative and demographic data and was first built in the early 90s. Even though it has passed through several iterations and improvements, the core is still very similar to its original one (based on Oracle 8). The second reason is an infrastructure called LIGHt which functions like a broker that is used to communicate health information from the state-provided software (i.e., SONHO) to third parties and is highly based on the HL7 version 2 standard [80]. This was a significant paradigm shift as this gateway provided the state-owned HIS the ability to connect with third parties. All this software and infrastructure is provided by SPMS, a state-owned enterprise, which is the governmental branch and the center of public eHealth policies, infrastructure, acting as regulator and HIS developer. This centralized top-down initiative was meant to enable better, faster interoperability progress, though it may come at the cost of innovation as private technology developers are unable to find a market in Portugal [81], and may explain why many large information technology companies do not have a significant presence in the country as Portugal's health information software systems are based on public, state-owned systems.

Due to the mandatory migration, virtually all public hospitals have the same basis for administrative data input and different health information systems for the clinical data (some public and some private). Private care delivery organizations are free to choose which health information software to use and now are not part of the HIE structure [82]. The data stored in public software is exchanged through the RSE ("Registo Saúde Eletrónico" – Portuguese for Electronic Health Record) [83]. From its inception, the RSE was designed to act as an aggregator of health data, with a similar structure to a hybrid federated approach having three different layers of information, ranging from local to shared/centralized data. In the middle layer, rests interoperability with existing systems. The catalog of data available at the moment in the RSE is difficult to collect and aggregate, but the high-level national goal is to provide access to general information about medication, diagnosis, demographic data, procedures, vaccination, and laboratory exams in the shared layer [84]. While the definition of full interoperability is constantly being updated, a major barrier to completion is physical hardware, where it is estimated that the public sector has between 25% and 50% of the necessary computer equipment across care delivery and public health agencies [85].

Future Considerations & Strategy: The current focus of the MoH is applying eHealth efforts supporting the patients, namely telemedicine and personal smartphone applications. Key points from Portugal's scene from 2021-2022 were the telemedicine support for its HIS. In 2019, the Portuguese MoH introduced PENTS (National Strategic Telehealth Plan), with an aim to provide access to telehealth across the country [86], with a focus on chronic illnesses [87]. Further, SNS24 – the personal app provided by SPMS has received several updates over the last two years, and can now be used by patients to access key health information due to connection with RSE. Vaccines, prescriptions, referrals, teleconsultations, allergies, and sick leaves are some of the possible information to view in the app [88]. These two points are defined by the national strategy for the Health Information Ecosystem 2020 which is focused on three aspects: governance and strategy alignment; improvement of human resources' competencies and technological upgrades [89, 90]. Finally, Portugal faces several future challenges which have yet to be addressed in a formal way. Data security is a growing issue, with several hospitals having been attacked by ransomware hackers

over the last few years and the strategy to respond to this matter is still being formalized [91]. Several workgroups were created, but a clear guideline or approach is yet to be determined. Further, patient consent processes remain an ongoing area of development in Portugal – currently, many patients engage in a two-step consent process where they consent for their inclusion and for each use of medical services for some features (such as e-prescribing), but for other features such as the use of HIE for clinical care by providers, implied consent (or an opt-out consent model) is the norm, while the patient-facing portals have stricter consent requirements [92]. Experts rated "Inadequate or inconsistent approaches to patient consent" as a major barrier to interoperability in Portugal in 2020 [93]. Additionally, data quality issues remain problematic – researchers have found large discrepancies in reconciling different electronic referral records in Portuguese hospitals [94]. Keeping in mind that evidence generated from data is the basis of clinical practice, ensuring quality and accuracy standards is key to going further into digital adoption and fulfillment of national strategies around the world.

4 Discussion

While deriving generalizable insights from the case studies of the health systems presented here is difficult, given the unique legislative and regulatory, policy, and health care delivery environments in each, our study contributes to our understanding of how policy shapes HIE adoption and use by identifying several commonalities across our five focal nations. First, successful national HIE requires some level of central strategic planning and involvement, irrespective of health system type. As Payne et al. noted in their paper comparing HIE internationally, "*in countries that have successfully achieved HIE, or are on course to do so, the impetus came from government and the change was galvanized with economic incentives to health care providers*"[7]. Despite operating within different payment and care delivery environments, both Israel and Portugal have more advanced HIE capabilities than the US, in part due to the emphasis placed on data exchange by a national-level policy. However, centralization is not a guarantee of complete success without adversity – the UK NHS has, and continues to, experienced challenges in their HIE strategy despite the extremely high level of organizational centralization in the UK. However, "bottom-up" innovation and early adoption can spur important progress, such as the adoption of early HIE systems by Israel's largest HMO, Clalit, which provided a blueprint for national adoption. The US may soon experience a similar process – despite the relatively "hands-off" nature of their early HIE strategy, several pockets of successful HIE did develop, which serve as a model for a new strategy following the 21st Century Cures Act focusing on APIs, FHIR standards, and penalties for information blocking.

The variation in HIE policy and success across varying nations reflects the multiple challenges of HIE, including aligning financial incentives for adoption and use, technical challenges, governance choices, and integration of outside data into clinical workflows. If the only serious impediment to broad interoperability were financial incentives, the UK NHS would have had a simple time achieving robust HIE. Similarly, while HIE is far from robust in the US, it is likely more advanced than the German system despite featuring an even more decentralized, competitive, and market-oriented care delivery system. Socio-technical challenges around setting standards, generating awareness of the HIE and encouraging adoption without burdening clinician and patient users, and building governance models with broad acceptance that are flexible enough to adapt to changing technological and social needs are common across our case study nations.

While there is no clear "best" overarching HIE strategy, multiple approaches may be effective in facilitating robust interoperable data exchange. For example, it may be that a "middle-out" system, defined by Price *et al.* as centralized leadership with strong public-private collaboration and more local control by individual care delivery organizations similar to the current strategy in the UK, may provide a balanced approach to these challenges, preserving local innovation around issues like workflow integration by private vendors and health care organizations while setting national-level interoperability incentives, standards, and governance frameworks [55]. Similarly, heavily centralized approaches may be successful, as Israel's increasing centralization has shown. However, the early experience of the US during the HITECH era illustrates the difficulty of achieving broad, robust interoperability with a heavily decentralized policy framework – even with significant effort to build EHR infrastructure and align financial incentives. Each system must recognize the importance and centrality of the patients – from giving them the final say in the ability to opt-out of data sharing such as in Israel to enabling them to access all their clinical data in the US, any national HIE strategy must work to ensure patients' changing needs are being met. Policymakers may wish to focus on creating integrated care structures that maximize incentives to share data while simultaneously ensuring there is a national-level interoperability strategy that makes data exchange a priority, with prescriptive guidance around standards and governance. At the same time, these strategies need to be adaptable to an ever-changing world, and researchers should carefully monitor and evaluate HIE progress to inform policymakers and allow them to be flexible in their approach. Finally, ensuring awareness among users (e.g. clinicians, patients, payers, public health agencies, etc.) and supporting actual use of HIE is critical to move from encouraging building interoperable data exchange to maximizing the potential of HIE to improve health and care delivery.

Directions for Future Study, Research and Policy Implications: While there is a significant and growing literature, there are still many unknowns with respect to HIE. While we provide speculation above regarding how different policy frameworks have influenced the development of HIE in five nations, in many dimensions of HIE we are unable to make direct cross-country comparisons due to a lack of data or published research. Standardized measures of HIE in terms of data exchanged, semantic interoperability, use by frontline clinicians are often not available at the national-level, or only available in aggregate form or with organization-level survey

measures that do not provide granular details of use. Our analysis was limited to holistic, qualitative evaluations of HIE maturity, centralization, and incentives due to this lack of broad, comparable empirical measures, and we supplemented what literature exists with small and single-site studies, policy briefs, and grey literature. To remedy this, we provide a non-exhaustive list of areas for additional future research:

- *Measurement*: Many measures of HIE focus on organization-level adoption, rather than evaluating actual real-world use (e.g., how often data from HIE is used in care delivery, or aggregated into population-level data by public health agencies), especially in studies measuring at a national-level [33, 95]. More granular measurement, including measuring volume of outside records exchanged, viewed, and their influence on the patient and clinician, is the next frontier of HIE measurement. Finally, measuring whether and how clinicians actually use that data when delivering care is critical to improving HIE policy at a national level and implementation at a local level.
- *Impact*: While early modeling studies have predicted significant cost reductions from HIE [3], many empirical studies have found it difficult to identify broad gains in either quality or cost [96]. Additional research, especially studies with robust designs for causal inference, is important to identify the impact of HIE on quality, costs, and the patient experience [97].
- *Workflow*: Given growing concerns over clinician EHR work [98, 99], especially in the US [100], it is critical for HIE to present outside records within existing workflows without exacerbating burden. Early studies show integrating HIE into the standard patient history screen greatly increased clinician views of outside records [101], future research should evaluate how and when to present information from various sources (e.g., other providers; patient reported data; remote monitoring devices) to clinicians across various clinical scenarios and settings (e.g., inpatient, outpatient, emergency; post-acute care; community health providers).
- *Population Health and HIE as "Data Utility":* While HIE has always held great potential as a data aggregation source to fuel population-level analytics, the COVID-19 pandemic demonstrated the importance of this information. Several efforts are underway to conceptualize HIEs more broadly as a "data utility", including using HIE data for more accurate clinical quality measurement, data aggregation for research, and public health analytics [102, 103]. While still a nascent field, following the development of these next-generation efforts and evaluating their effectiveness is critical to guide the next phase of HIE.

HIE research is at an important inflection point – moving past structural measures of adoption to evaluating volume and impact at scale, best practices for integration of data into clinical workflows across a range of scenarios and broadening the spectrum of HIE use cases. Scientific research plays a critical role in the continuous evaluation of HIE to inform the healthcare eco-system (including patients, policymakers and practitioners) of the value of data sharing and avoid potential pitfalls and unintended consequences.

5 Conclusion

Following the digitization of health care delivery, HIE and data interoperability has become increasingly important for delivering high quality care. HIE can reduce costs and duplicative utilization, improve the patient experience, and empower patients with access to their data, and ensure clinicians have access to all relevant information for their patients at the point of care, no matter where that data was generated. While implementing robust national HIE has been a challenge, our case studies highlighting different approaches from five nations suggest some generalizable policy frameworks that balance centralized decision-making around standards and governance but allow for local innovation in technical capability and workflow integration. Future research on HIE should include more detailed measurement of data exchange and use, the impact of interoperability on patients and care delivery organizations and assessing new frontiers of using HIE for public and population health.

References

1. Adler-Milstein J, Jha AK. HITECH Act drove large gains in hospital electronic health record adoption. Health Aff (Millwood)2017;36(8):1416–22. doi: 10.1377/hlthaff.2016.1651.
2. Blumenthal D. Stimulating the Adoption of Health Information Technology. N Engl J Med 2009;360(15):14779. doi: 10.1056/NEJMp0901592.
3. Walker J, Pan E, Johnston D, Adler-Milstein J, Bates DW, Middleton B. The value of Health care information exchange and interoperability. Health Aff (Millwood) 2005;Suppl Web Exclusives:W5-10-W5-18. doi: 10.1377/hlthaff.w5.10.
4. Kyle MA, Frakt AB. Patient administrative burden in the US health care system. Health Serv Res 2012;56(5):755–65. doi: 10.1111/1475-6773.13861.
5. Holmgren AJ, Apathy NC, Adler-Milstein J. Barriers to Hospital Electronic Public Health Reporting and Implications for the COVID-19 Pandemic. J Am Med Inform Assoc 2020;27(8):1306-9. doi: 10.1093/jamia/ocaa112.
6. Ammenwerth E, Duftschmid G, Al-Hamdan Z, Bawadi H, Cheung NT, Cho K-H, et al. International Comparison of Six Basic eHealth Indicators Across 14 Countries: An eHealth Benchmarking Study. Methods Inf Med 2020;59(S 02):e46–e63. doi: 10.1055/s-0040-1715796.
7. Payne JH, Lovis C, Gutteridge C, Pagliari C, Natarajan S, Yong C, et al. Status of health information exchange: a comparison of six countries. J Glob Health 2019;9(2):0204279. doi: 10.7189/jogh.09.020427.
8. Adler-Milstein J, Pfeifer E. Information Blocking: Is It Occurring and What Policy Strategies Can Address It? Milbank Q 2017;95(1):117–35. doi: 10.1111/1468-0009.12247.
9. Downing K, Mason J. ONC Targets Information Blocking. J AHIMA 2015;86(7):36–8.
10. Liyanage H, Krause P, de Lusignan S. Using ontologies to improve semantic interoperability in health data. J Innov Health Inform 2015;22(2). doi: 10.14236/jhi.v22i2.159.
11. de Mello BH, Rigo SJ, da Costa CA, da Rosa Righi R, Donida B, Rosecler Bez M, et al. Semantic interoperability in health records standards: a systematic literature review. Health Technol (Berl) 2022;12(2):255–72. doi: 10.1007/s12553-022-00639-w.
12. Akhlaq A, Sheikh A, Pagliari C. Defining health information exchange: scoping review of published definitions. J Innov Health Inform 2017;23(4) doi: 10.14236/jhi.v23i4.838.
13. Sarkar N. Transforming Health Data to Actionable Information: Recent Progress and Future Opportunities in Health Information Exchange. Yearb Med Inform 2022;31:203–14. doi: 10.1055/s-0042-1742519.
14. Dixon BE, Holmgren AJ, Adler-Milstein J, Grannis SJ. Health Information Exchange and Interoperability. In: Finnell JT, Dixon BE, editors. Clinical Informatics Study Guide: Text and Review. Cham: Springer International Publishing; 2022. p. 203–19. doi: 10.1007/978-3-030-93765-2_14.

15. Adler-Milstein J, Bates DW, Jha AK. A Survey of Health Information Exchange Organizations in the United States: Implications for Meaningful Use. Ann Intern Med 2011;154(10):666–71. doi: 10.1059/0003-4819-154-10-201105170-00006.
16. Adler-Milstein J, Garg A, Zhao W, Patel V. A Survey Of Health Information Exchange Organizations In Advance Of A Nationwide Connectivity Framework. Health Aff (Millwood) 2012;40(5):736–44. doi: 10.1377/hlthaff.2020.01497.
17. Gematik, Pressemitteilung | Fakten zur Digitalisierung: TI-Atlas 2022 veröffentlicht | gematik, Oct. 18, 2022. [available at: https://www.gematik.de/presse/pressemitteilung-fakten-zur-digitalisierung-ti-atlas-2022-veroeffentlicht; accessed Dec. 15, 2022].
18. Dehling T, Sunyaev A. German Health IT Infrastructure: A Large-Scale Network Alleviating Challenges in Health IT Development. Working Paper, University of Cologne, May 2014. doi: 10.13140/RG.2.1.3315.8566.
19. Holmgren A, Adler-Milstein J. Health Information Exchange in US Hospitals: The Current Landscape and a Path to Improved Information Sharing. J Hosp Med 2017;12(3):193. doi: 10.12788/jhm.2704.
20. Blumenthal D. Launching HITECH. N Engl J Med 2010;362(5):382–5. doi: 10.1056/NEJMp0912825.
21. Apathy NC, Holmgren AJ, Adler-Milstein J. A decade post-HITECH: Critical access hospitals have electronic health records but struggle to keep up with other advanced functions. J Am Med Inform Assoc 2012;28(9):1947-54. doi: 10.1093/jamia/ocab102.
22. 2020 Medicare Promoting Interoperability Program Scoring Methodology Fact Sheet, Centers for Medicare & Medicaid Services, 2020. [Available at: https://www.cms.gov/files/document/scoring-methodology-fact-sheet.pdf; accessed Mar. 15, 2023].
23. Medicare Access and CHIP Reauthorization Act of 2015 (MACRA). [Available at: https://www.congress.gov/114/plaws/publ10/PLAW-114publ10.pdf; accessed Mar. 29, 2023].
24. Grossman JM, Kushner KL, November EA. Creating sustainable local health information exchanges: can barriers to stakeholder participation be overcome? Res Brief 2008;2:1-12.
25. Charles D, Swain M, Patel V. Interoperability among U.S. Non-federal Acute Care Hospitals, 2014. ONC Data Brief 2015;25:10.
26. Office of the National Coordinator for Health Information Technology. 2022 Report to Congress: Update on the Access, Exchange, and Use of Electronic Health Information, 2022. [Available at: https://www.healthit.gov/sites/default/files/page/2023-02/2022_ONC_Report_to_Congress.pdf; accessed Mar. 29, 2023].
27. Holmgren AJ, Everson J, Adler-Milstein J. Association of Hospital Interoperable Data Sharing With Alternative Payment Model Participation. JAMA Health Forum 2002;3(2):e215199–e215199. doi: 10.1001/jamahealthforum.2021.5199.
28. Madhavan S, Bastarache L, Brown JS, Butte AJ, Dorr DA, Embi PJ, et al. Use of electronic health records to support a public health response to the COVID-19 pandemic in the United States: a perspective from 15 academic medical centers. J Am Med Inform Assoc 2012;28(2):393–401. doi: 10.1093/jamia/ocaa287.
29. Reeves JJ, Pageler NM, Wick EC, Melton GB, Gamaliel Tan Y-H, Clay BJ, et al. The Clinical Information Systems Response to the COVID-19 Pandemic. Yearb Med Inform 2012;30:105–25. doi: 10.1055/s-0041-1726513.
30. Holmgren AJ, Patel V, Charles D, Adler-Milstein J. US Hospital Engagement in Core Domains of Interoperability. Am J Manag Care 2016;22(12):e395–e402.
31. Holmgren AJ, Patel V, Adler-Milstein J. Progress In Interoperability: Measuring US Hospitals' Engagement In Sharing Patient Data. Health Aff (Millwood) 2017;36(10):1820–7. doi: 10.1377/hlthaff.2017.0546.
32. Apath, NC, Vest JR, Adler-Milstein J, Blackburn J, Dixon, BE, Harle CA. Practice and market factors associated with provider volume of health information exchange. J Am Med Inform Assoc 2021;28(7):1451–60. doi: 10.1093/jamia/ocab024.
33. Patel V, Pylypchuk Y, Parasrampuria S, Kachay L. Interoperability among Office-Based Physicians in 2015 and 2017. ONC Data Brief 2019;47:12.
34. Apathy NC, Holmgren AJ. Opt-in consent policies: potential barriers to hospital health information exchange. Am J Manag Care 2020;26(1):e14.
35. Bronsoler A, Doyle J, Schmit C, Van Reenen J. The Role of State Policy in Fostering Health Information Exchange in the United States. NEJM Catal Innov Care Deliv 2023;4(1). doi: 10.1056/CAT.22.0302.
36. Winden T, Boland L, Frey N, Satterlee P, Hokanson J. Care everywhere, a point-to-point HIE tool: Utilization and impact on patient care in the ED. Appl Clin Inform 2014;5(2):388–401. doi: 10.4338/ACI-2013-12-RA-0100.
37. Everson J, Butler E. Hospital adoption of multiple health information exchange approaches and information accessibility. J Am Med Inform Assoc:27(4):577–83. doi: 10.1093/jamia/ocaa003.
38. Cross DA, Adler-Milstein J. Investing in Post-Acute Care Transitions: Electronic Information Exchange Between Hospitals and Long-Term Care Facilities. J Am Med Dir Assoc 2017;18(1):30–4. doi: 10.1016/j.jamda.2016.07.024.
39. Lin SC, Hollingsworth JM, Adler-Milstein J. Alternative Payment Models and Hospital Engagement in Health Information Exchange. Am J Manag Care 2019;25(1):e1–e6.
40. Apathy NC, Holmgren AJ, Werner RM. Growth in health information exchange with ACO market penetration. Am J Manag Care 2022;28(1):e7–e13. doi: 10.37765/ajmc.2022.88815.
41. Porreca D, Yaraghi N. The Impact of Population Health Analytics on Health Care Quality and Efficacy Among CPC+ Participants. Milbank Memorial Fund; 2022. [Available at: https://www.milbank.org/wp-content/uploads/2022/10/WNY_Report2.pdf; accessed Mar. 29, 2023].
42. About ONC's Cures Act Final Rule. [Available at: https://www.healthit.gov/curesrule/overview/about-oncs-cures-act-final-rule; accessed Mar. 15, 2023].
43. Trusted Exchange Framework and Common Agreement | HealthIT.gov, Office of the National Coordinator for Health Information Technology, May 26, 2020. [Avaialable at: https://www.healthit.gov/topic/interoperability/trusted-exchange-framework-and-common-agreement; accessed Mar. 15, 2023].
44. ONC debuts first cohort of qualified networks in TEFCA, including Epic, eHealth Exchange, Healthcare Dive. [Available at: https://www.healthcaredive.com/news/onc-first-cohort-of-qualified-networks-tefca-epic/642654/; accessed Feb. 27, 2023].
45. Office of the National Coordinator for Health Information Technology, Trusted Exchange Framework and Common Agreement (TEFCA) | HealthIT.gov, HealthIT.gov. [Available at: https://www.healthit.gov/topic/interoperability/policy/trusted-exchange-framework-and-common-agreement-tefca; accessed Dec. 15, 2022].
46. Bogumil-Uçan S, Klenk T. Varieties of health care digitalization: Comparing advocacy coalitions in Austria and Germany. Rev Policy Res 2021;38(4):478–503. doi: 10.1111/ropr.12435.
47. Müller J, Ullrich C, Poss-Doering R. Beyond Known Barriers—Assessing Physician Perspectives and Attitudes Toward Introducing Open Health Records in Germany: Qualitative Study. J Particip Med 2020;12(4):e19093. doi: 10.2196/19093.
48. Naumann L, Babitsch B, Hübner UH. eHealth policy processes from the stakeholders' viewpoint: A qualitative comparison between Austria, Switzerland and Germany. Health Policy Technol 2021;10(2):100505. doi: 10.1016/j.hlpt.2021.100505.
49. Naumann L, Esdar M, Ammenwerth E, Baumberger D, Hübner U. Same Goals, Yet Different Outcomes: Analysing the Current State of eHealth Adoption and Policies in Austria, Germany, and Switzerland Using a Mixed Methods Approach. MEDINFO 2019; 2019. p. 1012–6. doi: 10.3233/SHTI190377.
50. Pohlmann S, Kunz A, Ose,D, Winkler EC, Brandner A, Poss-Doering R, et al. Digitalizing Health Services by Implementing a Personal Electronic Health Record in Germany: Qualitative Analysis of Fundamental Prerequisites From the Perspective of Selected Experts, J Med Internet Res 2020 Jan 29;22(1):e15102. doi: 10.2196/15102.
51. EHR and PHR: digital records in the German healthcare system, Mar. 18, 2019. [Available at: https://www.gesundheitsindustrie-bw.de/en/article/news/ehr-and-phr-digital-records-in-the-german-healthcare-system; accessed Feb. 28, 2023].
52. Amelung V, Angelkorte M, Augurzky B, Brauer R, Freigang F, Fritzsche F, et al. Ergebnisse der ersten nationalen Reifegradmessung deutscher Krankenhäuser. DigitalRadar 2022;148. [Available at: https://www.bundesgesundheitsministerium.de/service/publikationen/details/digitalradar-zwischenbericht.html; accessed Mar. 29, 2023].
53. Albrecht M, Otten M, Sander M, Temizdemir E, Wichtrup W. PraxisBarometer Digitalisierung

2020. Stand und Perspektiven der Digitalisierung in der vertragsärztlichen und -psychotherapeutischen Versorgung; 2020. [Available at: https://www.iges.com/sites/igesgroup/iges.de/myzms/content/e6/e1621/e10211/e24893/e26423/e26424/e26426/attr_objs26428/IGES_KBV_Digitalbaromenter_2020_ger.pdf; accessed Mar. 29, 2023].
54. Thiel R, Deimel L, Schmidtmann D, Piesche K, Hüsing T, Rennoch J, et al. Gesundheitssystem-Vergleich Fokus Digitalisierung. #SmartHealthSystems Digitalisierungsstrategien im internationalen Vergleich; 2018. [Available at: https://www.bertelsmann-stiftung.de/fileadmin/files/Projekte/Der_digitale_Patient/VV_SHS-Gesamtstudie_dt.pdf; accessed Mar. 29, 2023].
55. Price C, Green W, Suhomlinova O. Twenty-five years of national health IT: exploring strategy, structure, and systems in the English NHS. J Am Med Inform Assoc 2019;26(3):188–97. doi: 10.1093/jamia/ocy162.
56. NHS England - Transformation Directorate. Consent and confidential patient information; Feb. 10, 2023. [Available at: https://transform.england.nhs.uk/information-governance/guidance/consent-and-confidential-patient-information/; accessed Feb. 27, 2023].
57. Justinia T. The UK's National Programme for IT: Why was it dismantled? Health Serv Manage Res 2017;30(1):2–9. doi: 10.1177/0951484816662492.
58. Warren LR, Clarke J, Arora S, Darzi A. Improving data sharing between acute hospitals in England: an overview of health record system distribution and retrospective observational analysis of inter-hospital transitions of care. BMJ Open 2019;9(12):e031637. doi: 10.1136/bmjopen-2019-031637.
59. NHS England Interoperability. [Available at: https://www.england.nhs.uk/digitaltechnology/connecteddigitalsystems/interoperability/; accessed Feb. 28, 2023].
60. NHS England - Transformation Directorate. Information Governance Framework: Shared Care Records; Feb. 02, 2022. [Available at: https://transform.england.nhs.uk/information-governance/guidance/summary-of-information-governance-framework-shared-care-records/; accessed Dec. 30, 2022].
61. Zhang J, Sood H, Harrison OT, Horner B, Sharma N, Budhdeo S. Interoperability in NHS hospitals must be improved: the Care Quality Commission should be a key actor in this process. J R Soc Med 2020;113(3):101–4. doi: 10.1177/0141076819894664.
62. Watkinson F, Dharmayat KI, Mastellos N. A mixed-method service evaluation of health information exchange in England: technology acceptance and barriers and facilitators to adoption. BMC Health Serv Res 2012;21(1):737. doi: 10.1186/s12913-021-06771-z.
63. Wise J. New data strategy for England aims to rebuild public trust. BMJ 2022;377:o1455. doi: 10.1136/bmj.o1455.
64. Topol E. The topol review. Preparing the healthcare workforce to deliver the digital future; Feb 2019. p. 1–48.
65. Scobie S, Castle-Clarke S. Implementing learning health systems in the UK NHS: Policy actions to improve collaboration and transparency and support innovation and better use of analytics. Learn Health Syst 2020;4(1):e10209. doi: 10.1002/lrh2.10209.
66. Marjanovic S, Altenhofer M, Hocking L, Chataway J, Ling T. Innovating for improved healthcare: Sociotechnical and innovation systems perspectives and lessons from the NHS. Sci Public Policy 2020;47(2):283–97. doi: 10.1093/scipol/scaa005.
67. Frankel M, Chinitz D, Salzberg CA, Reichman K. Sustainable health information exchanges: the role of institutional factors. Isr J Health Policy Res 2013;2(1):21. doi: 10.1186/2045-4015-2-21.
68. Flaks-Manov N, Shadmi E, Hoshen M, Balicer RD. Health information exchange systems and length of stay in readmissions to a different hospital. J Hosp Med 2016;11(6):401–6. doi: 10.1002/jhm.2535.
69. Politi L, Codish S, Sagy I, Fink L. Balancing volume and duration of information consumption by physicians: The case of health information exchange in critical care. J Biomed Inform 2017 Jul;71:1–15. doi: 10.1016/j.jbi.2017.05.007.
70. Nirel N, Rosen B, Sharon A, Blondheim O, Sherf M, Cohen AD. OFEK virtual medical records: an evaluation of an integrated hospital-community system. Harefuah 2011 Feb;150(2):72–8, 209.
71. Gilboa A. The Implementation of a National Health Information Exchange Platform in Israel. Inter-American Development Bank; May 2022. doi: 10.18235/0004272.
72. Colombo F, Oderkirk J, Slawomirski L. Health Information Systems, Electronic Medical Records, and Big Data in Global Healthcare: Progress and Challenges in OECD Countries. In: Haring R, Kickbusch I, Ganten D, Moeti M, editors. Handbook of Global Health. Cham: Springer International Publishing; 2020. p. 1–31. doi: 10.1007/978-3-030-05325-3_71-1.
73. State of Israel Ministry of Health. Health Information Exchange. [Available at: https://www.health.gov.il/English/About/projects/shared_medical_info/Pages/default.asp; accessed Dec. 15, 2022].
74. Zimlichman E. Israel: A Nationwide Health Information Exchange Program in Israel: A Unique Case Study. In: Health Systems Improvement Across the Globe. CRC Press; 2018.
75. World Health Organization, Israel. Advancing Interoperability and Data Sharing in the Health System; Jan. 2022. [Available at: https://www.euro.who.int/__data/assets/pdf_file/0008/509597/vignette-interoperability-data-sharing-Israel.pdf; accessed Feb. 27, 2023].
76. Itzhaki M, Koren E, Abu Hussein K, Levy L, Gantz I, Barnoy S. Use of Health Information Exchange in the Continuity of Care as Viewed by Patients and Nurses: A Cross-Sectional Study. CIN Comput Inform Nurs 2020 Mar;:10.1097/CIN.0000000000000861. doi: 10.1097/CIN.0000000000000861.
77. Balicer R, Shadmi E, Manor O, Leventer-Roberts M. Israel: Structural and Functional Integration at the Israeli Healthcare System. In: Amelung V, Stein V, Suter E, Goodwin N, Nolte E, Balicer R, editors. Handbook Integrated Care. Cham: Springer International Publishing; 2021. p. 1055–63. doi: 10.1007/978-3-030-69262-9_62.
78. Ben-David R. Israel approves 19 programs for $30 million digital health initiative. The Times of Israel; May 30, 2022. [Available at: https://www.timesofisrael.com/israel-approves-19-programs-for-30-million-digital-health-initiative/; accessed Dec. 15, 2022].
79. Dispatch n.o 182/2013 of Ministry of Health, 182; Sep. 2013. [Available at: https://dre.pt/dre/detalhe/despacho/12071-2013-2529406; accessed Dec. 15, 2022].
80. Interoperabilidade Técnica: LIGHt; PNB; NCP, SPMS, 2017. [Available at: https://www.spms.min-saude.pt/2017/06/interoperabilidade-tecnica-light-pnb-ncp/; accessed Dec. 15, 2022].
81. Catan G, Espanha R, Mendes RV, Toren O, Chinitz D. Health information technology implementation - impacts and policy considerations: a comparison between Israel and Portugal. Isr J Health Policy Res 2015 Aug;4(1):41. doi: 10.1186/s13584-015-0040-9.
82. Mendonça SR. The roadmap to healthcare digitalization: factors that affect the Portuguese approach to ehealth; 2022. [Available at: https://run.unl.pt/handle/10362/144871; accessed Dec. 14, 2022].
83. ACSS, RSE – Registo de Saúde Electrónico – R2A: Orientações para Especificação Funcional e Técnica do Sistema de RSE, 2009. [Avaialable at: https://www.essnortecvp.pt/upload/files/R2A_Orientacoes_FuncTec_v3.pdf; accessed Dec. 15, 2022].
84. Registo de Saúde Eletrónico, SPMS. [Available at: https://www.spms.min-saude.pt/2020/07/registo-de-saude-eletronico/; accessed Dec. 15, 2022].
85. Simões JA, Augusto GF, Fronteira I, Hernández-Quevedo C. Portugal: health system review. Health Syst Transit 2011;19(2):1-184.
86. Martins H, Monteiro M, Loureiro P, Cortes M. SPMS - Serviços Partilhados do Ministério da Saúde. E.P.E. Centro Nacional de TeleSaúde Plano Estratégico Nacional para a Telessaúde 2019-2022, 201. [Available at: https://www.spms.min-saude.pt/; accessed Mar. 29, 2023].
87. Canhão H. Portugal in the e-health path. EIT Health 2020 Feb 28. [Available at: https://eithealth.eu/news-article/portugal-launches-its-national-telehealth-as-the-first-country-in-the-world/; accessed Dec. 15, 2022].
88. SPMS. Aplicação móvel SNS 24, 2022. [Available at: https://www.sns24.gov.pt/guia/utilizar-a-aplicacao-movel-sns-24/; accessed Dec. 15, 2022].
89. Resolução do Conselho de Ministros n.o 62/2016 | DRE," 67, Oct. 2016. [Available at: https://dre.pt/dre/detalhe/resolucao-conselho-ministros/62-2016-75542124; accessed Dec. 15, 2022].
90. SPMS. ENESIS 2020 – 2022; 2019. [Available at: https://www.spms.min-saude.pt/wp-content/uploads/2019/10/ENESIS2022_VersaoParaConsultaPublicaOut2019.pdf; accessed Dec. 15, 2022].
91. Portugal: Cyberattack on Hospital do Divino Espírito Santo Impacting Notification of COVID-19 Test Results; 2021 Jun 28. [Available

at: https://www.databreaches.net/portugal-cyber-attack-on-hospital-do-divino-espirito-santo-impacting-notification-of-covid-19-test-results/; accessed Jan. 02, 2023].
92. Gonçalves ME, Raimundo J. Over Troubled Water: E-Health Platforms and the Protection of Personal Data: The Case of Portugal. Port J Public Health 2017;35(1):52–66. doi: 10.1159/000477650.
93. Rucker D, Hasan A, Lewis L, Tao D. GDHP-Advancing Interoperability Together Globally. Global Digital Health Partnership, Sydney, Australia. GDHP White Paper on Interoperability 2020 Jul. [Available at: https://www.healthit.gov/sites/default/files/page/2021-01/GDHP-Advancing%20Interoperability%20Together%20Globally.pdf; accessed Feb. 28, 2023].
94. Vasco Santos J, Martins FS, Lopes F, Souza J, Freitas A. Discharge status of the patient: evaluating hospital data quality with a focus on long-term and palliative care patient data. Health Inf Manag J 2021 Nov30;18333583211054160. doi: 10.1177/18333583211054161.
95. Savage M, Neinstein A, Adler-Milstein J. Measure The Impact Of The ONC's New Interoperability Rules Now. Health Affairs Blog; 2020 Jul 7.
96. Rahurkar S, Vest JR, Menachemi N. Despite The Spread Of Health Information Exchange. There Is Little Evidence Of Its Impact On Cost, Use, And Quality Of Care. Health Aff (Millwood) 2015 Mar;34(3):477–83. doi: 10.1377/hlthaff.2014.0729.
97. Menachemi N, Rahurkar S, Harle CA, Vest JR. The benefits of health information exchange: an updated systematic review. J Am Med Inform Assoc 2018 Sep;25(9):1259–65. doi: 10.1093/jamia/ocy035.
98. Adler-Milstein J, Zhao W, Willard-Grace R, Knox M, Grumbac, K. Electronic health records and burnout: Time spent on the electronic health record after hours and message volume associated with exhaustion but not with cynicism among primary care clinicians. J Am Med Inform Assoc 2020;27(4):531–8. doi: 10.1093/jamia/ocz220.
99. Overhage JM, McCallie D. Physician Time Spent Using the Electronic Health Record During Outpatient Encounters: A Descriptive Study. Ann Intern Med 2020 Feb 4;172(3):169-74. doi: 10.7326/M18-3684.
100. Holmgren AJ, Downing NL, Bates DW, Shanafelt TD, Milstein A, Sharp CD, et al. Assessment of Electronic Health Record Use Between US and Non-US Health Systems. JAMA Intern Med 2021 Feb 1;181(2):251-9. doi: 10.1001/jamainternmed.2020.7071.
101. Adler-Milstein J, Wang MD. The impact of transitioning from availability of outside records within electronic health records to integration of local and outside records within electronic health records. J Am Med Inform Assoc 2020 Apr;27(4):606–12. doi: 10.1093/jamia/ocaa006.
102. D'Amore JD, McCrary LK, Denson J, Li C, Vitale CJ, Tokachichu P, et al. Clinical data sharing improves quality measurement and patient safety. J Am Med Inform Assoc 2021 Jul14;28(7):1534–42. doi: 10.1093/jamia/ocab039.
103. Indiana Network for Population Health focuses on social determinants of health, EurekAlert! [Available at: https://www.eurekalert.org/news-releases/653368; accessed Dec. 15, 2022].
104. Esdar M, J. Hüsers J, Weiß JP, Rauch J, Hübner U. Diffusion dynamics of electronic health records: A longitudinal observational study comparing data from hospitals in Germany and the United States. Int J Med Inform 2019 Nov;131:103952. doi: 10.1016/j.ijmedinf.2019.103952.
105. Nohl-Deryk P, Brinkmann JK, Gerlach FM, Schreyögg J, Achelrod D. [Barriers to Digitalisation of Healthcare in Germany: A Survey of Experts], Gesundheitswesen Bundesverb. Arzte Offentlichen Gesundheitsdienstes Ger 2018 Nov;80(11):939–45. doi: 10.1055/s-0043-121010.
106. Payne TH, Detmer DE, Wyatt JC, Buchan IE. National-scale clinical information exchange in the United Kingdom: lessons for the United States. J Am Med Inform Assoc 2011;18(1):91–8. doi: 10.1136/jamia.2010.005611.
107. Warren L, Clarke J, Darzi A. Measuring the Scale of Hospital Health Record System Fragmentation in England. Health Serv Res 2020;55(S1):43–4. doi: 10.1111/1475-6773.13386.
108. NHS. A plan for digital health and social care. National Health Service; Jun. 2022. [Available at: https://www.gov.uk/government/publications/a-plan-for-digital-health-and-social-care/a-plan-for-digital-health-and-social-care; accessed Feb. 28, 2023].
109. Lejbkowicz I, Denekamp Y, Reis S, Goldenberg D. Electronic medical record systems in Israel's public hospitals. Isr Med Assoc J IMAJ 2004 Oct;6(10):583–7.
110. Heart T, O'Reilly P, Sammon D, O'Donoghue J. Bottom-up or top-down? A comparative analysis of electronic health record diffusion in Ireland and Israel. J Syst Inf Technol 2009 Jan;11(3):244–68. doi: 10.1108/13287260910983623.
111. Prey JE, Polubriaginof F, Kuperman JG, Tiase V, Collins SA, Vawdrey DK. International perspectives on sharing clinical data with patients. Int J Med Inform 2016 Feb;86:135–41. doi: 10.1016/j.ijmedinf.2015.11.007.
112. Hübner UH, Wilson GM, Morawski TS, Ball MJ. Nursing Informatics: A Health Informatics, Interprofessional and Global Perspective. Springer Nature; 2022. doi: 10.1007/978-3-030-91237-6.
113. Vest JR. Health information exchange: national and international approaches. Adv Health Care Manag 2012;12:3–24. doi: 10.1108/s1474-8231(2012)0000012005.
114 Grenha Teixeira J, de Pinho NF, Patrício L. Bringing service design to the development of health information systems: The case of the Portuguese national electronic health record. Int J Med Inf Inform 2019 Dec;132:103942. doi: 10.1016/j.ijmedinf.2019.08.002.

Correspondence to:
A Jay Holmgren, PhD MHI
10 Koret Way, Office 327A
San Francisco, CA 94131
USA
Tel: +1 517 896 0614
E-mail: a.holmgren@ucsf.edu